Date: _____ Day: _____

Chapter 2 (Bluman)
Freq. Distribution & Graphs

2.1 - Organizing Data

Example : 2.1

. Question From Book (Buluman chapter 2 Page 37)

. min value in Data : 37 (Age)
max value in Data : 90 (Age)

1. Find range :

∴ Range = Max value - Min value
Range = 90 - 37
= 53

2. Decide no of classes.

no of classes = C = 1 + 3.3 log (n)
= 1 + 3.3 log (50)
= 1 + 3.3 (1.7)
= 1 + 5.61
= 6.61 ≈ 7

Day:

Date:

3. Find Width:

$\therefore$ Width = Range / no of classes = 53/7 = 7.5 (round off.) = 8

Lower class Bdry = 36 − 0.5 = 35.5

Upper class Bdry = 43 + 0.5 = 43.5

Frequency Distribution Table

Class Limits	Class Boudaries	Mid Point (x_i)	TALLY	f
36 − 43	35.5 − 43.5	$\frac{36+43}{2} = 39.5$	𝍸	5
44 − 51	43.5 − 51.5	$\frac{44+51}{2} = 47.5$	III	3
52 − 59	51.5 − 59.5	$\frac{52+59}{2} = 55.5$	𝍸 III	8
60 − 67	59.5 − 67.5	$\frac{60+67}{2} = 63.5$	𝍸 III	8
68 − 75	67.5 − 75.5	$\frac{68+75}{2} = 71.5$	𝍸 𝍸 I	12
76 − 83	75.5 − 83.5	$\frac{76+83}{2} = 79.5$	𝍸 𝍸 II	11
84 − 91	83.5 − 91.5	$\frac{84+91}{2} = 87.5$	𝍸	5
[illegible]	[illegible]	[illegible]	[illegible]	50

Count how many Digits under this boundary

Date: _____ Day: _____

Relative Frequency:

$$\therefore R.f = \frac{\text{observed Freq}}{\text{Total Freq}}$$

Commulative Frequency

obtained by adding frequencies of all preceeding class including that class

Commulative Relative Frequency

$$\therefore C.R.f = \frac{Cf}{\text{Total Freq}}$$

class Bdries	f	R.f	C.f	C.R.f
35.43.5	5	$5/50 = 0.10$	5	$5/50 = 0.10$
49.5-51.5	3	$3/50 = 0.06$	$5+3=8$	$8/50 = 0.16$
51.5-59.5	8	$8/50 = 0.16$	$8+8=16$	$16/50 = 0.32$
59.5-67.5	8	$8/50 = 0.16$	$16+8=24$	$24/50 = 0.48$
67.5-75.5	10	$10/50 = 0.20$	$24+10=34$	$34/50 = 0.68$
75.5-83.5	11	$11/50 = 0.22$	$34+11=45$	$45/50 = 0.90$
83.5-91.5	5	$5/50 = 0.10$	$45+5=50$	$50/50 = 1.00$
Total	50	1.00	50	1.00

Date: _____ Day: _____

Example : 2.1

Distribution of Blood Types.

Blood Data.

A	B	B	AB	O
O	O	B	AB	B
B	B	O	A	O
A	O	O	O	AB
AB	A	O	B	A

Construct Frequency Distr: Table

Class	Tally	Freq.	Percent
A	𝍸	5	20
B	𝍸 II	7	28
O	𝍸 IIII	9	36
AB	IIII	4	16
		25	100

$\therefore \% f = \frac{\text{observed Freq.} \times 100}{\text{Total.}}$

Date: _____ Day: _____

Example: 2.3

MPGs for SUVs.

12	17	12	14	16	18
16	18	12	16	17	15
15	16	12	15	16	16
12	14	15	12	15	15
19	13	16	18	16	14

min value: 12

max value: 19

1. Range:

$\therefore R = max - min$

$R = 19 - 12$

$R = 7$ → Range is very small so we use ungrouped Dist.

2. No of class:

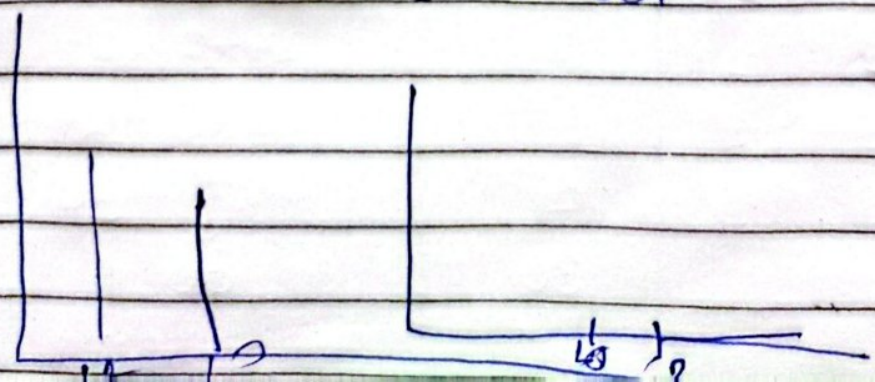
No of classes $= 1 + 3.3(\log n)$

$= 1 + 3.3 \log(30)$

$= 1 + 3.3(1.47)$

$= 1 + 4.851 \approx$

$= 5.851$



Date: _____ Day: _____

class Limits	TALLY	class Boundaries	Frequency
12	𝍸 I	11·5 - 12·5	6
13	I	12·5 - 13·5	1
14	III	13·5 - 14·5	3
15	𝍸 I	14·5 - 15·5	6
16	𝍸 III	15·5 - 16·5	8
17	II	16·5 - 17·5	2
18	III	17·5 - 18·5	3
19	I	18·5 - 19·5	1
			30

↓

must alway equal to no. of class.

class Bdries	Freq	Commulative Freq.
11·5 - 12·5	6	6
12·5 - 13·5	1	6+1 = 7
13·5 - 14·5	3	7+3 = 10
14·5 - 15·5	6	10+6 = 16
15·5 - 16·5	8	16+8 = 24
16·5 - 17·5	2	24+2 = 26
17·5 - 18·5	3	26+3 = 29
18·5 - 19·5	1	29+1 = 30
Σ	30	30

Date: _____ Day: _____

Example: 2.2

Record High Temprature

same as 1st Example (2.1)
Question on Pg # 41 (Bluman ch2)

Date: _____ Day: _____

Presentation of Quantitative Data

1. Freq. Distribution Table

1. Range:

$\therefore$ Range = max marks - min marks

$= 98 - 69$

$= 29$

2. No of classes (C):

$\therefore C = 1 + 3.3 \log(n)$

$= 1 + 3.3 \log(35)$

$= 6.09$

$= 7$

3. Class width (W)

$\therefore W = R/C = \frac{29}{7}$

$W = 4.14$

$= 5$

Date: _____ Day: _____

4. Freq. Distribution Table.

Frequence Distribution Table					
class Limits	Class Boundaries	Mid point (x_i) $\frac{68+72}{2} = 70$	Tally Mark	Frequency (f)	% Freq $= \frac{f}{\text{Total Freq}} \times 100$
68 – 72	67.5 – 72.5	70	I	1	2.86 %
73 – 77	72.5 – 77.5	75	I	1	2.86
78 – 82	77.5 – 82.5	80	I	1	2.86
83 – 87	82.5 – 87.5	85	IIII IIII	10	28.57
88 – 92	87.5 – 92.5	90	IIII IIII	10	28.57
93 – 97	92.5 – 97.5	95	IIII IIII I	11	31.42
98 – 102	97.5 – 102.5	100	I	1	2.86
			Σ	35	100 %

Date: _____ Day: _____

5. Relative Frequency

$$\therefore \text{r.f} = \frac{\text{observed Freq}}{\text{Total Freq}}$$

6. Commulative Frequency

obtained by adding Frequencies of all preceeding class including current class

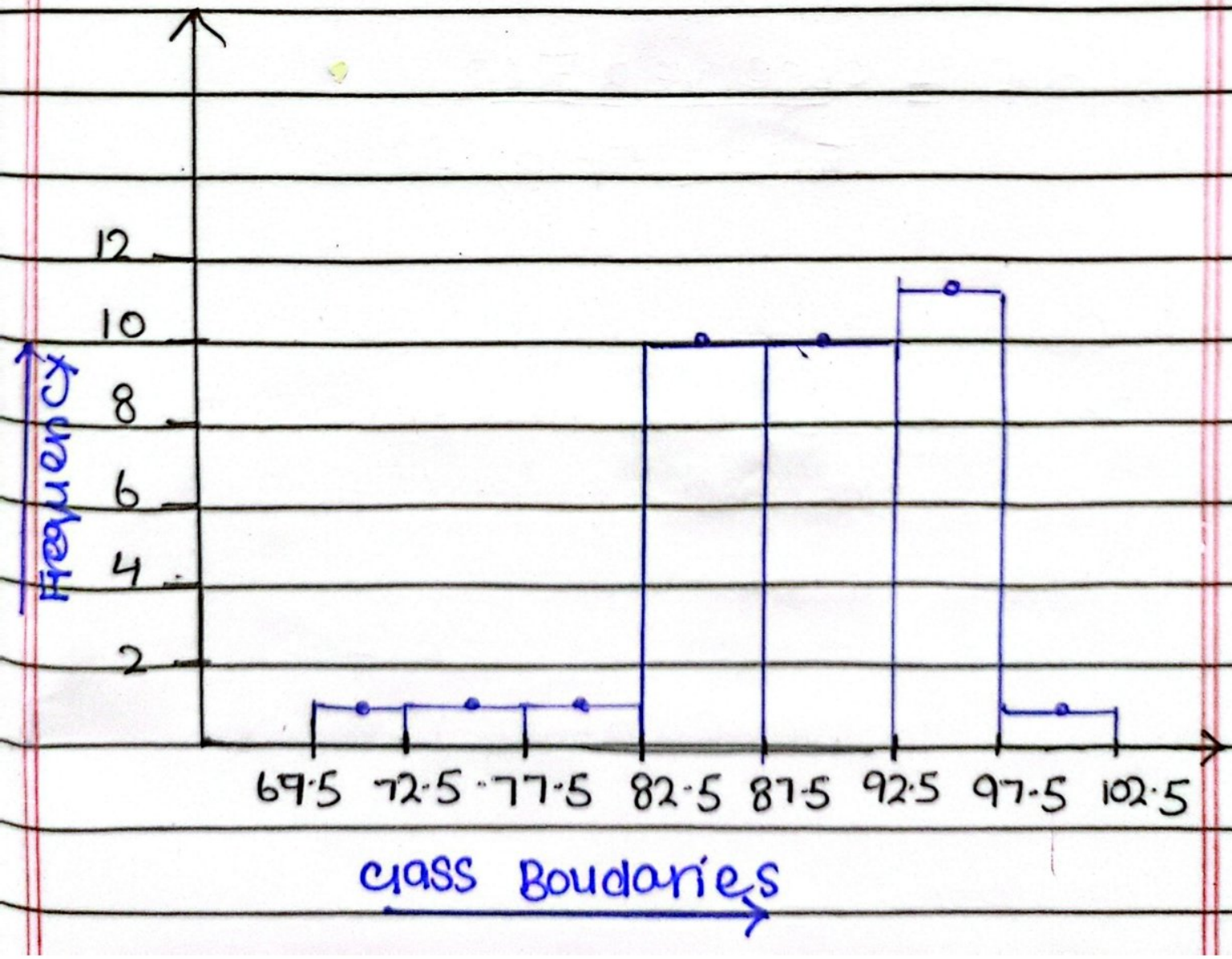
7. Commulative Relative Frequency

$$\therefore \text{C.r.f} = \frac{\text{C.f}}{\text{Total Freq}}$$

class Boundaries	f	Relative f	c.f	c.r.f
67.5 72.5	1	$\frac{1}{35}=0.0286$	1	$\frac{1}{35}=0.028$
72.5-77.5	1	0.0286	1+1=2	0.057
77.5-82.5	1	0.0286	2+1=3	0.085
82.5-87.5	10	0.2857	3+10=B	0.371
87.5-92.5	10	0.2857	13+10=23	0.657
92.5-97.5	11	0.3143	23+11=34	0.971
97.5-102.5	1	0.0286	34+1=35	1.000
Σ	35	1.0000	35	-

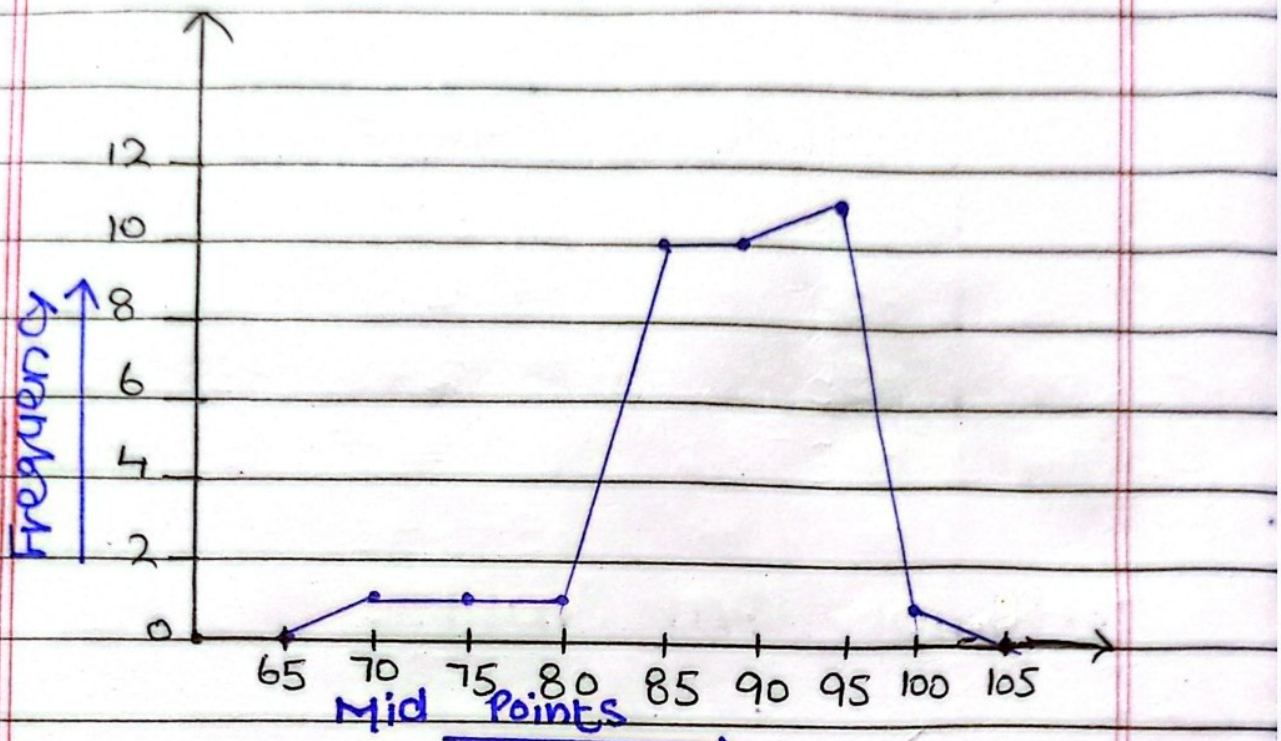
Date: _____ Day: _____

2. Histogram

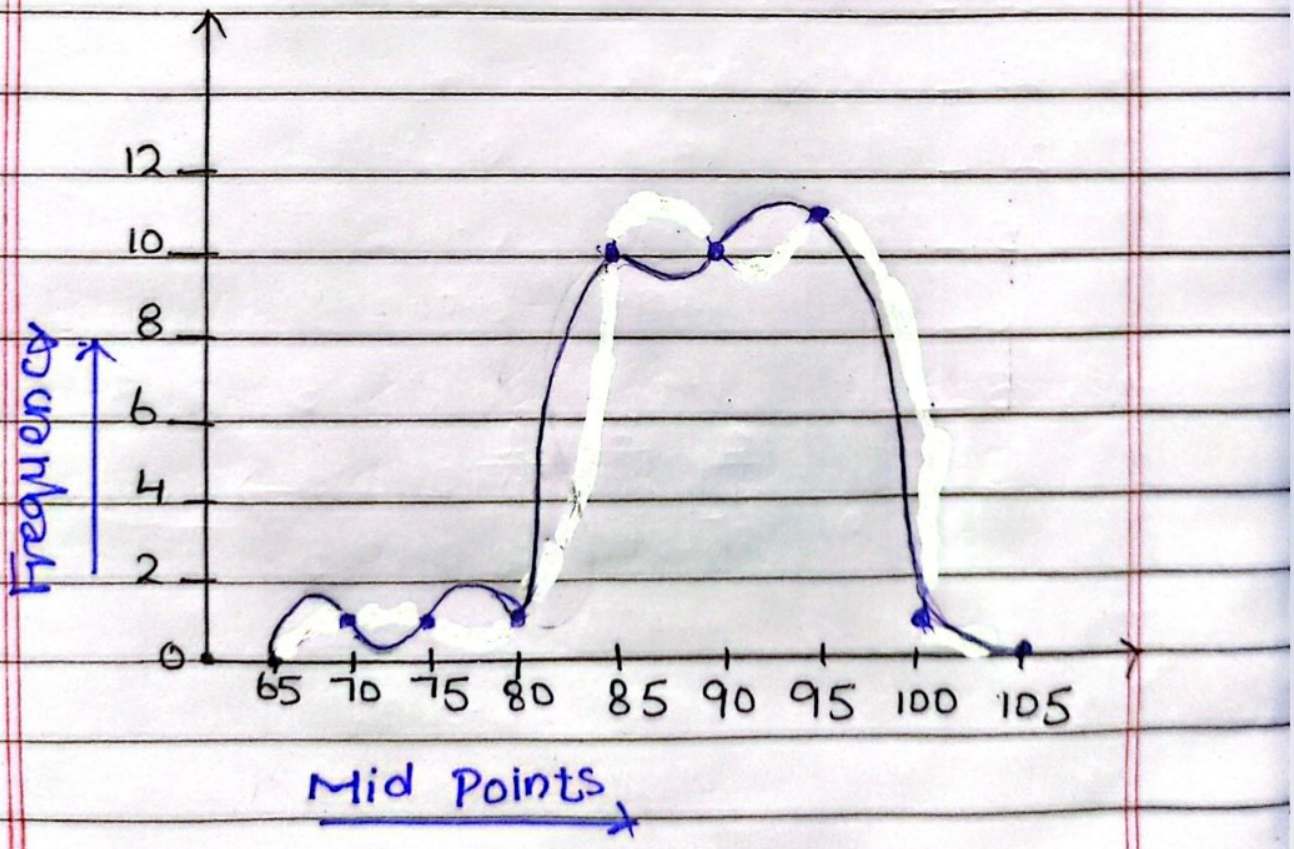


Date: _____ Day: _____

3. Frequency Polygon



4. Frequency Curve

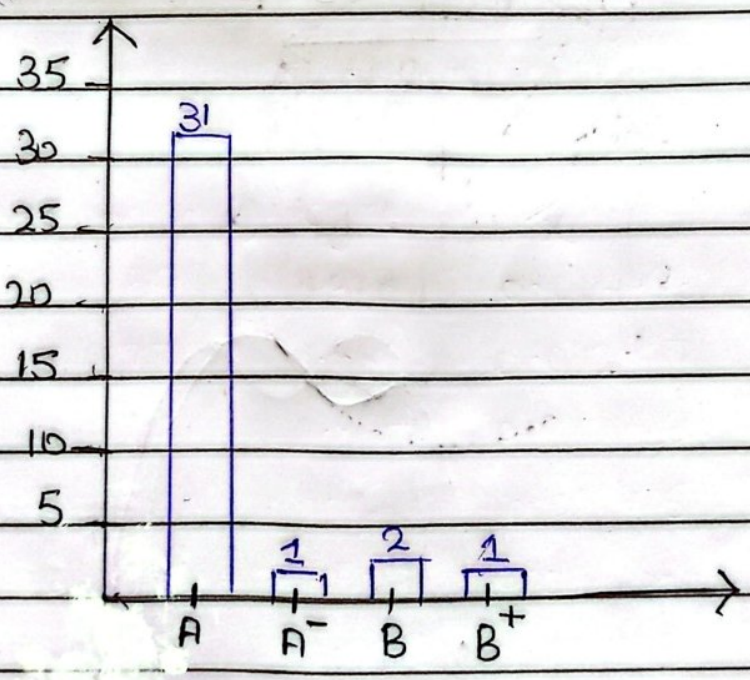


Date: _____ Day: _____

Presentation of Qualitative Data

Grade	Frequency
A	31
A^-	1
B	2
B^-	1

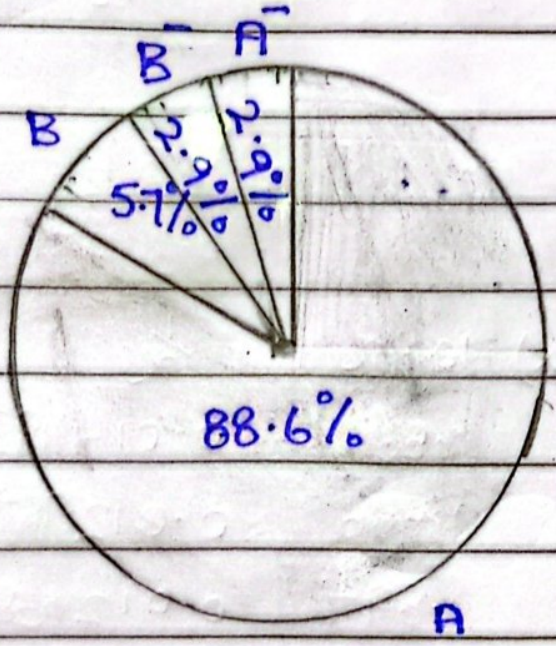
1. Simple Bar Chart



Date: _____ Day: _____

2. Pie Chart

Grade	Freq	% F	Degree
A	31	88.6	318.9°
A^-	1	2.9	10.3°
B	2	5.7	20.6°
B^-	1	2.9	10.3°
Σ	35	–	–



Date: _____ Day: _____

Measure of Central Tendency

Marks in Ascending order

69	85	87	89	91	93	95
77	85	87	89	92	93	96
79	86	87	89	92	95	96
84	86	88	85	93	95	97
85	86	89	90	93	95	98

Find sum of x, $\log x$, & $\frac{1}{x}$

Sr No	Mark (x)	log x	$\frac{1}{x}$
1	69	1.838	0.014493
2	77	1.886	0.012987
3	79	1.897	0.012658
4	84	1.924	0.011905
5	85	1.929	0.011765
6	85	1.929	0.011765
7	85	1.929	0.011765
8	86	1.934	0.011628
9	86	1.934	0.011628
10	86	1.934	0.011628
11	87	1.939	0.011494
12	87	1.939	0.011494
13	87	1.939	0.011494
14	88	1.944	0.011364

Date: _____ Day: _____

15	89	1·949	0·011236
16	89	1·949	0·011236
17	89	1·949	0·011236
18	89	1·949	0·011236
19	89	1·949	0·011236
20	90	1·954	0·011111
21	91	1·959	0·010989
22	92	1·963	0·010870
23	92	1·963	0·010870
24	93	1·968	0·010753
25	93	1·968	0·010753
26	93	1·968	0·010753
27	93	1·968	0·010753
28	95	1·977	0·010526
29	95	1·977	0·010526
30	95	1·977	0·010526
31	95	1·977	0·010526
32	96	1·982	0·010417
33	96	1·982	0·010417
34	97	1·986	0·010309
35	98	1·991	0·010204
Σ	3120	68·2000	0·394549

1. Arithmetic Mean

$$\therefore AM = \frac{\Sigma x}{n} = \frac{3120}{35}$$

$$\boxed{AM = 89.14} = \bar{x}$$

Date: Day:

2. Geometric Mean

$$\therefore GM = \text{antilog}\left(\frac{\sum \log x}{n}\right)$$

$$GM = \text{antilog}\left(\frac{68.2000}{35}\right)$$

$$GM = \text{antilog}(1.9486)$$

$$\boxed{GM = 88.83}$$

3. Or Harmonic Mean

$$\therefore HM = \frac{n}{\sum(1/x)} = \frac{35}{0.394549}$$

$$\boxed{HM = 88.71}$$

4. Median

$$\therefore \text{Position} = \left(\frac{n+1}{2}\right)^{th}.$$

$$P = \left(\frac{35+1}{2}\right)^{th} = 18$$

↳ Median Lies at 18 idx

$$\boxed{M = 89}$$

Date: Day:

5. Mod

∴ Mod = most Frequent value

Mod = 89

Measure of Dispersion

Sr	Marks (x)	$\lvert x-\mu \rvert$	$(x-\mu)^2$
1	69	$\lvert 69-89.14 \rvert = 20.14$	405.6196
2	77	$\lvert 77-89.14 \rvert = 12.14$	147.3796
3	79	$\lvert 79-89.14 \rvert = 10.14$	102.8196
4	84	$\lvert 84-89.14 \rvert = 5.14$	26.4196
5	85	$\lvert 85-89.14 \rvert = 4.14$	17.1396
6	85	$\lvert 85-89.14 \rvert = 4.14$	17.1396
7	85	$\lvert 85-89.14 \rvert = 4.14$	17.1396
8	86	$\lvert 86-89.14 \rvert = 3.14$	9.8596
9	86	$\lvert 86-89.14 \rvert = 3.14$	9.8596
10	86	$\lvert 86-89.14 \rvert = 3.14$	9.8596
11	87	$\lvert 87-89.14 \rvert = 2.14$	4.5796
12	87	$\lvert 87-89.14 \rvert = 2.14$	4.5796
13	87	$\lvert 87-89.14 \rvert = 2.14$	4.5796
14	88	$\lvert 88-89.14 \rvert = 1.14$	1.2996
15	89	$\lvert 89-89.14 \rvert = 0.14$	0.0196
16	89	$\lvert 89-89.14 \rvert = 0.14$	0.0196
17	89	$\lvert 89-89.14 \rvert = 0.14$	0.0196
18	89	$\lvert 89-89.14 \rvert = 0.14$	0.0196
19	89	$\lvert 89-89.14 \rvert = 0.14$	0.0196
20	90	$\lvert 90-89.14 \rvert = 0.86$	0.7396

Date: _____ Day: _____

21	91	$\|91-89.14\| = 1.86$	3.4596
22	92	$\|92-89.14\| = 2.86$	8.1796
23	92	$\|92-89.14\| = 2.86$	8.1796
24	93	$\|93-89.14\| = 3.86$	14.8996
25	93	$\|93-89.14\| = 3.86$	14.8996
26	93	$\|93-89.14\| = 3.86$	14.8996
27	93	$\|93-89.14\| = 3.86$	14.8996
28	95	$\|95-89.14\| = 5.86$	34.3396
29	95	$\|95-89.14\| = 5.86$	34.3396
30	95	$\|95-89.14\| = 5.86$	34.3396
31	95	$\|95-89.14\| = 5.86$	34.3396
32	96	$\|96-89.14\| = 6.86$	47.0596
33	96	$\|96-89.14\| = 6.86$	47.0596
34	97	$\|97-89.14\| = 7.86$	61.7796
35	98	$\|98-89.14\| = 8.86$	78.4996
Σ	3120	155.42	1230.2860

Absolute Measure of Dispersion

1. Range :-

$\therefore$ Range = max marks − min marks

Range = 98 − 69

Range = 29

Date: Day:

2. Mean Deviation

$$\therefore MD = \frac{\sum_{i=1}^{N} |x-\mu|}{N} = \frac{155.42}{35}$$

$$\boxed{MD = 4.44}$$

3. Standard Deviation

$$\therefore SD = \sigma = \sqrt{\frac{\sum_{i=1}^{N} (x-\mu)^2}{N}}$$

$$\sigma = \sqrt{\frac{1230.2860}{35}} = \sqrt{35.15}$$

$$\boxed{\sigma = 5.93}$$

4. Variance

$$\therefore \sigma^2 = \frac{\sum_{i=1}^{N} (x-\mu)^2}{35}$$

$$\boxed{\sigma^2 = 35.15}$$

Date: _____ Day: _____

Relative Measure of Dispersion

1. Coefficient of Range

$$\therefore C_R = \frac{X_{max} - X_{min}}{X_{max} + X_{min}} = \frac{98-69}{98+69}$$

$$\boxed{C_R = 0.17}$$

2. Coefficient of MD

$$\therefore C_{MD} = \frac{\mu MD}{\mu} = \frac{4.44}{89.14}$$

$$\boxed{C_{MD} = 0.05}$$

3. Coefficient of Variation

$$\therefore C_V = \frac{\sigma}{\mu} \times 100 = \frac{5.93}{89.14} \times 100$$

$$\boxed{C_V = 6.65\%}$$

Date: Day:

Quartiles

$\therefore$ Position of $Q_k = \frac{k(n+1)}{4}$

where $k \in \{1, 2, 3\}$

1. Quartile (Q_1)

Position of $Q_1 = \frac{1(35+1)}{4} = \frac{36}{4}$

$P = 9 \rightarrow Q_1$ lies at idx 9

$\boxed{Q_1 = 86}$

2. Quartile (Q_2)

$\therefore P = \frac{k(n+1)}{4} = \frac{2(35+1)}{4} = \frac{72}{4}$

$P = 18 \rightarrow Q_2$ lies at idx 18

$\boxed{Q_2 = 89}$

3. Quartile (Q_3)

$P = \frac{3(35+1)}{4} = \frac{108}{4} = 27 \rightarrow Q_3$ lies at idx 27

$\boxed{Q_3 = 93}$

Date: Day:

Percentile:

$\therefore$ Position of $P_k = \frac{k(n+1)}{100}$

where $k \in \{1, 2, 3, \ldots, 99\}$

1. Percentile (P_{40})

$$\text{Position} = \frac{k(n+1)}{100} = \frac{40(35+1)}{100} = \frac{1440}{100}$$

$P = 14.4 \rightarrow P_{40}$ lies b/w observation 14 & 15 at 14.4^{th}

$P_{40} = 88 + 0.4(89-88)$

$P_{40} = 88 + 0.4(1)$

$P_{40} = 88 + 0.4$

$\boxed{P_{40} = 88.4}$

Deciles:

$$\text{Position} = \frac{k(n+1)}{10}$$

where $k \in \{1, 2, \ldots, 9\}$

1. Decile (D_3)

$$P = \frac{k(n+1)}{10} = \frac{3(35+1)}{10} = \frac{108}{10}$$

Date: Day:

$P = 10.8 \rightarrow D_3$ Lies b/w observation

10 & 11

$D_3 = 86 + 0.8(87-86)$

$D_3 = 86 + 0.8(1)$

$D_3 = 86 + 0.8$

$\boxed{D_3 = 86.8}$

2. Decile (D_9)

$$P = \frac{k(n+1)}{10} = \frac{9(35+1)}{10} = \frac{324}{10}$$

$P = 32.4 \rightarrow D_9$ Lies b/w observation

32 & 33

$D_9 = 96 + 0.4(96-96)$

$D_9 = 96 + 0.4(0)$

$\boxed{D_9 = 96}$

Inter Quartile Range (IQR)

· IQR (Inter Quartile Range)

$\therefore IQR = Q_3 - Q_1$

$IQR = 93 - 86$

$\boxed{IQR = 7}$

Date: Day:

Box Plots

1. Lower Limit

$\therefore LL = Q_1 - 1.5(IQR)$

$LL = 86 - 1.5(7)$

$LL = 86 - 10.5$

$LL = 75.5$

2. Upper Limit

$\therefore UL = Q_3 + 1.5(IQR)$

$UL = 93 + 1.5(7)$

$UL = 93 + 10.5$

$UL = 103.5$

3. Outlier:

Any value less than Lower Limit & Greater than Upper Limit is outlier.

So Outlier : 69

Date: _____ Day: _____

4. Box Plot

